

Mujtaba Hussain Mirza

PhD Candidate in Computer Science | Sapienza University of Rome

📍 Rome, Italy | [Website](#) | [Google Scholar](#) | [LinkedIn](#) | [GitHub](#)

Research Profile

Final-year PhD candidate in Computer Science at Sapienza University of Rome, supervised by [Prof. Iacopo Masi](#) at the [OmnAI Lab](#). My research seeks to advance the **security, safety, robustness, and trustworthiness** of AI systems. Within these broader areas, I work on adversarial robustness, machine unlearning, test-time adaptation, model hallucinations, and multimedia forensics.

Education

PhD in Computer Science

Sapienza University of Rome

Supervisor: Prof. Iacopo Masi

Research Area: Adversarial machine learning.

Nov 2023 – Present

Rome, Italy

MSc in Computer Science

Sapienza University of Rome

Grade: 110/110 **cum laude**

Thesis: *Probing the Energy Landscape of Discriminative Classifiers with Adversarial Perturbations.*

Sep 2021 – Oct 2023

Rome, Italy

BTech in Computer Science and Engineering

University of Kashmir

CGPA: 8.01/10.

Thesis: *Stock Price Movement Prediction Using Market Indicators and Twitter Sentiment Analysis.*

Sep 2015 – Dec 2019

Kashmir, India

Publications

- **M. H. Mirza**, M. R. Briglia, S. Beadini, and I. Masi. “Shedding More Light on Robust Classifiers under the Lens of Energy-Based Models.” *European Conference on Computer Vision (ECCV)*, 2024.
- **M. H. Mirza**, A. D’Orazio, O. Melamed, and I. Masi. “A Provable Energy-Guided Test-Time Defense Boosting Adversarial Robustness of Large Vision-Language Models.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- M. R. Briglia, **M. H. Mirza**, G. Lisanti, and I. Masi. “Why Adversarially Train Diffusion Models?” *International Conference on Learning Representations (ICLR)*, 2026.
- G. Daidone, F. Bartolucci, M. R. Briglia, **M. H. Mirza**, G. Lisanti, and I. Masi. “Neutralizing Proactive Defense using Diffusion-based Upsampling.” *ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec)*, 2026.
- L. Mautino, **M. H. Mirza**, and I. Masi. “Correlating Adversarial Robustness with the Energy of Untrained Classifiers.” *International Conference on Machine Learning and Cybernetics (ICMLC)*, 2026.

Professional Experience

Associate Machine Learning Engineer

Jocata Financial Advisory & Technology

Dec 2020 – Dec 2021

Hyderabad, India

- Developed machine learning solutions for anti-money-laundering applications, improving model accuracy from 88% to 94%; contributed to MLOps adoption through MLflow and graph-based analysis with Neo4j.

Junior Research Analyst

Great Learning

Apr 2020 – Dec 2020

Bangalore, India

- Developed technical content on machine learning and Python for the Great Learning Blog and Great Learning Academy.

Academic Service and Activities

Workshop

Co-organizer, Workshop and Challenge on Unlearning and Model Editing (U&ME), ECCV 2026. [\[Link\]](#)

Organization

Peer Review

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions on Information Forensics and Security (TIFS), The Visual Computer (TVC).

Student Supervision

Co-supervised a BSc thesis on artistic style erasure in Stable Diffusion using machine unlearning techniques.

Research

Presentations

Selected for oral presentation at IEEE-EURASIP Summer School on Signal Processing (S3P 2024), Capri, Italy; presented a poster at CISP-ELLIS Summer School 2025.

Conference Service

Volunteer, International Conference on Image Analysis and Processing (ICIAP), Italy.

Technical Skills

Programming & ML

Python, C/C++, PyTorch, Keras, scikit-learn, Hugging Face, OpenCV, NumPy, Pandas, and PySpark.

Tools & Systems

MLflow, Git, Docker, Linux, AWS, SQL, PostgreSQL, Neo4j, and Jupyter.

Awards & Funding

- Sapienza University PhD Scholarship, awarded by Sapienza University of Rome through competitive selection (2023).
- ELSA (European Lighthouse on Secure and Safe AI) Travel Grant for participation in the CISP-ELLIS Summer School (2025).

Interests

Beyond research, I enjoy blogging (my first paid job), traveling (a habit I picked up after moving to Europe), cooking (where my experiments usually end in a delicious dinner), and hiking (perhaps a consequence of growing up in the mountains of Kashmir).